THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · EASY LEVEL CHAPTER TEST SET 2

Chapter 6: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The reverse of 1208 is: 1

(a) 8021 **(b)** 2081 **(c)** 0128 **(d)** 1208

2. Which of these numbers is a palindrome? 1

(a) 4545 **(b)** 4455 **(c)** 4564 **(d)** 4554

3. Which of these numbers is divisible by 9? 1

(a) 4564 **(b)** 4562 **(c)** 4563 **(d)** 4565

4. In the Collatz rule (halve if even, multiply by 3 and add 1 if odd), the number after 12 is: 1

(a) 6 **(b)** 7 **(c)** 8 **(d)** 5

5. The sum of the first 9 odd numbers is: 1

(a) 82 **(b)** 83 **(c)** 81 **(d)** 80

6. A two-digit number has tens digit 3 and units digit 9. Its value is: 1

(a) 39 **(b)** 38 **(c)** 41 **(d)** 40

7. The product of two odd numbers is: 1

(a) a multiple of 2 **(b)** odd **(c)** even **(d)** zero

8. The sum of the digits of 2906 is: 1

(a) 19 **(b)** 18 **(c)** 16 **(d)** 17

9. The reverse of 9051 is: 1

(a) 519 **(b)** 1509 **(c)** 9051 **(d)** 0159

10. Which of these numbers is a palindrome? 1

(a) 27827 **(b)** 27782 **(c)** 27872 **(d)** 28772

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Use reverse-and-add on 48 until a palindrome appears. 2

12. Without dividing, test whether 5136 is divisible by 3 and by 9. 2

13. Write the Collatz sequence that starts from 10 (halve if even; triple and add 1 if odd) until you reach 1. 2

14. Use reverse-and-add on 35 until a palindrome appears. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Apply one step of Kaprekar's routine to 4271 (largest arrangement of the digits minus smallest arrangement). 3

16. The sum of three consecutive numbers is 102. Find the numbers. 3

17. Without calculating, decide whether $15 \times 11 + 8$ is even or odd, and give a reason. 3

OR

Apply one step of Kaprekar's routine to 6309 (largest arrangement of the digits minus smallest arrangement). 3

18. The sum of three consecutive numbers is 123. Find the numbers. 3

OR

Without calculating, decide whether $8 \times 6 + 9$ is even or odd, and give a reason. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Take the two-digit number 38. Add it to its reverse, show that the sum is divisible by 11 and find the quotient. 5

OR

Write the Collatz sequence starting from 12. Find the number of steps to reach 1, the largest number reached and how many even numbers occur. 5

20. Take the two-digit number 59. Add it to its reverse, show that the sum is divisible by 11 and find the quotient. 5

OR

Write the Collatz sequence starting from 13. Find the number of steps to reach 1, the largest number reached and how many even numbers occur. 5

END OF QUESTION PAPER — Total: 40 Marks

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